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DATA DESCRIPTOR

A multimodal stress detection dataset with facial expressions and physiological signals

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Affective computing has garnered the attention and interest of researchers in recent years, as there is a need for AI systems to better understand and react to human emotions. However, analyzing human emotions, such as mood or stress, is quite complex. While various stress studies use facial expressions and wearables, most existing datasets rely on processing data from a single modality. This paper presents EmpathicSchool, a novel dataset that captures facial expressions and the associated physiological signals, such as heart rate, electrodermal activity, and skin temperature, under different stress levels. The data was collected from 30 participants during different sessions for about ninety minutes each (for a total of 40 hours). The data includes seven different signal types, including both computer vision and physiological features that can be used to detect stress. In addition, various experiments were conducted to validate the signal quality.

Background & Summary

Affective computing is an interdisciplinary field aimed at developing systems that can identify, recognize, and interpret human emotions¹. Affective computing leverages physiological signals to detect the individual's emotions or state of stress. Stress is one of the major problems in modern society that leads to significant health expenditure². Early ambulatory stress detection has become an exciting challenge^{3,4} because early stress detection can mitigate the conversion of stress to a chronic disease² and prevent stress-related health issues⁵. Human emotions manifest through physiological, behavioral, and cognitive changes, which can be utilized to adapt digital environments. For example, the current state of a smartphone user can be continuously computed based on their interactions with various digital services⁶, allowing personalized services to be provided in response to the user's mental state⁷. Additionally, stress factors, such as high workload, lack of autonomy, and long working hours, can negatively impact people's health⁸. Many studies point out that prolonged exposure to stress leads to chronic conditions, such as obesity⁹ or hypertension¹⁰, which may exacerbate conditions, such as type-II diabetes¹¹. Therefore, monitoring and understanding stress in workplaces is essential, especially in professions with increased exposure to stress, often leading to burnout and increased turnover¹².

Stress is primarily a physiological response to a stimulus, typically triggered by an external factor in an environment^{13,14}. Epinephrine, commonly known as adrenaline, is a significant stress hormone¹⁵. In a daily routine, a moderate amount of stress is considered beneficial¹⁶. It causes gentle excitement and improves performance when carrying out regular daily tasks. A reasonable amount of stress that causes excitement is termed eustress or positive stress^{17,18}. On the other hand, undesirable stress is called distress or negative stress^{18,19}. Repeated stress events can lead to emotional exhaustion²⁰, and the resulting long-term physiological stress can be harmful; such stress is termed chronic stress²¹. The effects of chronic stress may include anxiety, depression, sleep disturbances, and weight gain²². It may even compromise immune responses, increasing vulnerability to several diseases²³. Studies related to stress have gained interest in recent years due to its widespread effects on health, family, workplace productivity, society, and the economy²⁴.

The analysis of stress involves using wearable sensors to deduce physiological signals. Table 1 shows the standard signals provided by wearables. In contrast, the study and analysis of human emotions use a camera to infer facial expressions that represent emotions. In the literature, emotions and stress signals are typically considered two disjoint topics and are usually studied independently. Schmidt *et al.*²⁵ identify the gap in analyzing basic

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Signal	Device	Abbr
Heart Activity	Electrocardiography	ECG EKG
Skin Response	Electrodermal Activity	EDA
Skin Response	Galvanic Skin Response	GSR
Muscle Activity	Electromyography	EMG
Respiratory Response	Electromagnetic Generation	RR
Respiratory Response	Respiratory Inductive Plethysmograph	RIP
Blood Volume Pulse	Cardiovascular Dynamics	BVP HR HRV
Skin Temperature	Thermistor	TEMP
Brain Activity	Electroencephalogram	EEG
Eye Activity	Corneo-retinal Standing	EOG
Physical Activity	3-axis Accelerometer	ACC
Physical Activity	3-axis Magnetometer	MGM
Heart Activity	Sphygmomanometer	BP
Eye Characteristics	Pupil Diameter Detector Module	PD
Body Temperature	Thermal Imaging	TI

Table 1. Common physiological signals.

emotions and stress together; however, there is still a need for a dataset that addresses the issue of correlating stress and facial expressions under different stressful situations. This work presents a multimodal stress-emotion dataset containing stress data from wearable devices and emotion data extracted from facial expressions through a video feed. We analyze human expressions and physiological signals under different stress levels. The dataset presented is the first effort toward identifying stress from non-wearable devices in general and facial expressions in particular. The current study is inspired by our previous work on stress detection of nurses using wearable devices²⁶ and our initial effort to use facial expressions to study satisfaction²⁷. This study collected stress and facial expression data from 30 participants under different stress levels.

There is a host of surveys in the stress research literature. Alberdi *et al.*²⁸ listed physiological signals used for stress detection. Fukazawa *et al.*²⁹ introduced standard stress detection tools and devices. Carneiro³⁰ described techniques to access and monitor stress in offices and other workplaces. Alberdi *et al.*³¹ provided a comprehensive survey on stress recognition for office environments. They described the signals and their related features, as well as successful neural network models that detect stress. Using wearables and smartphones², investigated stress in daily life. They briefly described stress and its impact on society and investigated the stress tests and related signals. However, they did not discuss instruments and devices detecting stress, signal properties, machine learning features, or training parameters. In another paper, Fukazawa *et al.*³² provided a comprehensive survey of stress detection signals and techniques from a machine learning perspective. They only used location, activity, phone usage, context, sleep, and speech features to detect stress. They covered several papers that used a specific signal for stress detection and investigated how commonly a method or signal is used by showing its using percentage across the years. Panicker *et al.*³³ presented a comprehensive survey on stress detection. They provided a comprehensive description of emotions and their organismic subsystems. They also provided definitions of stress, enumerated stress detection signals and devices, presented three different stages of stress, and evaluated the correlation and differences between different types of stress and emotions. Zhang *et al.*³⁴ explored the integration of multiple physiological signals, including EEG, EMG, GSR, and ECG, for emotion detection across various daily activities. By constructing a deep neural network architecture, they achieved 95 percent accuracy in binary classification of arousal by combining task-specific representations for each signal type. The proposed LSTM-based model demonstrated high accuracy, emphasizing the complementary nature of physiological and visual stress indicators. Kurniawan³⁵ investigated several classification methods, such as SVM and Gaussian Mixture Models (GMM), to detect stress levels using GSR and speech features. They gained between 70 to 80 percent accuracy using GSR and 92 percent accuracy by combining both the speech and GSR using four classification techniques, namely K-means, decision tree, GMM, and SVM classification models. Moreover, some authors use thermal images to detect stress or deception detection³⁶.

The AffectiveRoad dataset³⁷ contains physiological signals to study the stress level of 10 drivers while driving in various environments. The study was conducted with drivers taking an 86-minute driving test in Tunisia. Schmidt *et al.*²⁵ introduced the WESAD (Wearable Stress and Affect Detection) dataset and studied the stress of 15 students while watching a movie and taking a TSST (Trier Social Stress Test) test³⁸ using E4 signals. MDPSPD (multimodal dataset for psychological stress detection)³⁹ is a multimodal stress detection dataset of EDA and PPG signals collected from university students while performing different tests (e.g., TSST⁴⁰, IQ test, and color-word tests⁴¹). Mundnich *et al.*⁴² provided TILES-2018, a multi-sensor dataset that provides a battery of surveys to cover personality traits, behavioral states, job performance, and well-being over time. The SWELL⁴³ dataset provides data corresponding to different stress levels of participants while performing some office work (e.g., answering email) at three different stress levels (neutral, stressor, and stressor with interruption) using Kinect, ECG, and emotional expressions extracted from videos. However, this dataset does not provide the videos of the participants and uses outdated facial features to detect emotion and stress. In addition, the SWELL dataset provides the stress level of the entire task as one label. Zaman *et al.*⁴⁴ studied the impact of email interruptions and office activities on productivity and stress, using multimodal data from 63 workers who were

exposed to batch or continual email interruptions, with or without additional stress. The data collected, including physiological stress indicators, writing quality measures, keystroke dynamics, and participant profiles, is expected to provide insights into personalized email management strategies and a better understanding of office activity dynamics.

Although WESAD²⁵ and SWELL⁴³ have been invaluable for stress detection research, they have some limitations. WESAD provides synchronized ECG, EDA, and motion data, but no facial video or expression labels. This makes WESAD single-modal from the vision standpoint and unable to capture behavioral cues. SWELL adds facial-expression features, but only reports them as coarse, single-label summaries per task (neutral vs. stressed), does not provide the raw video data, and relies on feature extractors, preventing reanalysis or application of new deep learning methods. Furthermore, both datasets offer workload labels at session-level granularity, which limits its analysis of rapid stress fluctuations. As a result, there is a need for a multimodal, high-resolution, open dataset that couples raw facial video and physiological streams with minute-level stress annotations.

Our dataset, EmpathicSchool, provides physiological stress signals streams collected from Empatica E4 and facial features from videos of engineering and computer science students performing different tasks. Our primary motivation for creating this dataset was to generate a multimodal emotion-stress detection dataset and analyze students' emotions and stress levels while performing routine tasks such as preparing for a presentation or taking an exam. In addition, we analyze the emotions and stress levels of the students after task fulfillment. To summarize, EmpathicSchool addresses the gaps observed in the existing datasets in four principal ways:

- **True multimodality:** we release both raw facial-video 1080p at 30 fps frame rate and synchronized physiological data (EDA, BVP, ACC, HR, and TEMP), enabling joint vision-biometric signal modeling.
- **Fine-grained labels:** self-assessment stress and NASA-TLX workload scores are provided at 2-minute intervals, allowing detection of transient stress changes, unlike the task-level single labels in SWELL.
- **High temporal resolution:** all modalities are timestamp-aligned, and video frames and biometric samples can be precisely upsampled or downsampled to any common rate up to 30 Hz, supporting both frame-based and windowed analysis.
- **Open, extensible data:** we publish raw videos, landmark files, raw E4 streams, and all code for feature extraction and preprocessing so new facial-expression or biometric signal feature extraction algorithms can be applied directly.

By combining these improvements, EmpathicSchool substantially broadens the scope for research in near-real-time, multimodal stress detection.

Methods

This section briefly introduces the sensors and protocols used in the data collection process. We also discuss the methods used to extract features from the video data and provide a summary of the sensors used for biomedical data acquisition.

The data was collected in two locations: Tampere University, Finland, and the University of Louisiana at Lafayette, USA. We received approval from the institutional review boards of the University of Louisiana at Lafayette (SP-21-144-IRI) and the Tampere region's ethics committee (50/2021) for the study's protocol for the corresponding locations. Before data collection, each participant received an IRB-approved presentation detailing the nature and duration of the experimental tasks, the exact signals to be captured by the Empatica E4 wristband (including electrodermal activity, blood-volume pulse, skin temperature, and tri-axial acceleration) and a laptop camera, and the intention to publish de-identified physiological time-series with either their videos or facial-landmark data in an open repository, with a suggested consent revocation window of two months after data collection. For the data collected at Tampere University, participants may request the withdrawal of their data at any time prior to the publication of the paper. After publication, withdrawal is no longer possible, as all personally identifiable information is permanently removed and the data is fully anonymized. A written informed consent was then obtained in two stages. The main consent form authorized data collection and explicitly granted permission for public dissemination of the de-identified dataset, while preserving the participant's right to withdraw at any point before release. Immediately after the experiment, a separate debriefing form required participants to choose whether they would (a) share the de-identified facial-feature data only or (b) share both the video recordings and facial-feature data; this second document was also signed by all the participants in this dataset. All personal identifiers are stored separately under lock and key (consent forms) and on an encrypted server (video data) for six months after the data collection and before destruction, ensuring compliance with the confidentiality safeguards mandated by both ethics committees. No waiver of consent was requested from the ethical review board.

Participants. The participants were recruited via flyers and recruitment emails in the engineering and computer science departments of the University of Louisiana at Lafayette and via email at Tampere University. We recruited 31 students, and after the data collection, 1 subject asked to be removed from the dataset. As a result, we had data from 30 participants aged 21 to 35 (Mean=25.3, Standard-Deviation=4.3). We asked participants (S7-S30) in Louisiana not to drink alcohol or coffee 24 hours before data collection. No such restrictions were applied to participants S1-S6 in Tampere. The exclusion criteria were pregnancy, heavy smoking, mental disorders, and chronic or cardiovascular diseases. Data collection in the United States was conducted in two phases to achieve the suggested minimum of 30 subjects for studies involving physiological and behavioral data⁴⁵.

Video data. We asked the participants to perform all the tasks in a sitting position, and the camera captured only their faces and shoulders. The videos of the participants were collected using a 1080p external webcam



Fig. 1 Participants' sitting and camera positions. *This image is captured from one of the authors who consented to have his image in the paper; we are not providing any of the participants' images in the manuscript.

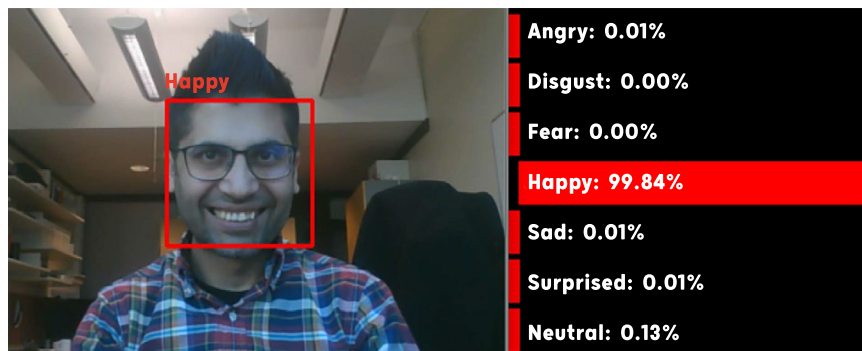


Fig. 2 The deployed system captures and displays emotions from live video frames, using the Haar cascade algorithm for face detection and the MiniXception model for facial expression recognition. *This image is captured from one of the authors who consented to have his image in the paper; we are not providing any of the participants' images in the manuscript.

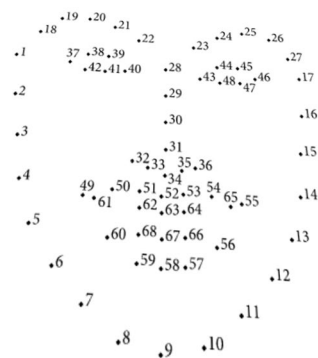
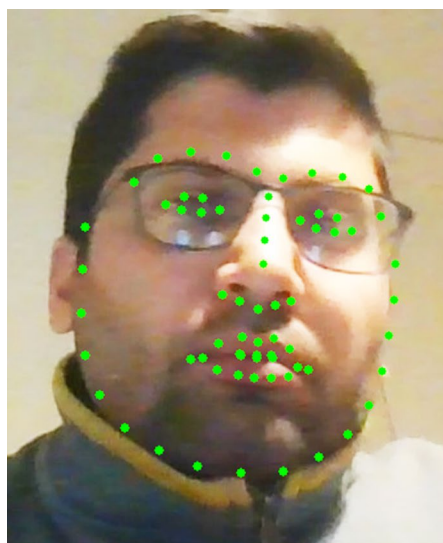


Fig. 3 68 facial landmarks. *This image is captured from one of the authors who consented to have his image in the paper; we are not providing any of the participants' images in the manuscript.

Signal	Abbreviation	Frequency
Electrodermal Activity	EDA	4.0 Hz
Heart Rate	HR	1.0 Hz
Skin Temperature	ST	1.0 Hz
Accelerometer	ACC	32 Hz
Inter-Beat Interval	IBI	—
Blood Volume Pulse	BVP	64 Hz

Table 2. Signals and the sampling frequency of Empatica E4 sensors.

mounted on top of a laptop screen with 30 frames per second. As approved by the institutional review board at the University of Louisiana at Lafayette, the video data of the participants who consented to release their video files are provided as part of the dataset. We post-processed the recorded videos to extract facial expressions as discussed in the following subsections. Figure 1 illustrates the experimental setup and the camera’s position.

Facial expression recognition. The facial expression recognition system consists of two modules. In the first module, we used the Haar cascade⁴⁶ frontal face detection algorithm implemented in the OpenCV⁴⁷ library to detect the face in the video frame. Haar cascade uses Haar-like features to encode the local appearance of faces⁴⁸. The detected faces were processed and passed through a pre-trained model to recognize facial expressions in the second module. We used MiniXception⁴⁹, trained over the Facial Expression Recognition 2013 (FER-2013) dataset⁵⁰ for facial expression recognition. FER-2013 dataset contains 28,709 training images, 3,589 validation images, and 3,589 test images from the seven basic facial expression categories (0=Angry, 1=Disgust, 2=Fear, 3=Happy, 4=Sad, 5=Surprise, 6=Neutral). Our experiments deduced the facial expressions from all the video recordings. Figure 2 demonstrates real-time emotion display through the utilization of the Haar cascade algorithm for face detection in conjunction with the MiniXception model for facial expression recognition. These data are provided as part of the dataset.

Dlib features. Facial landmarks are used in various computer vision tasks, such as drowsiness detection⁵¹, fatigue detection⁵², facial expression recognition⁵³, and micro-expression classification⁵⁴. Therefore, we provide the facial landmarks of all the video frames collected as part of the dataset. We utilize Dlib⁵⁵, which returns the position of 68 facial landmarks. An example of landmark detection on a facial image based on Dlib is shown in Fig. 3.

Facial Action Units. Facial Action Coding System (FACS) is a comprehensive framework for understanding facial expressions, particularly the discrete movements and gestures referred to as Action Units (AUs). Recent studies have leveraged the granularity of the FACS to analyze stress levels through facial expressions^{56–59}. We do not provide pre-computed AUs in the dataset. However, we provide the code to generate Facial Action Unit features from facial landmarks features in our GitHub repository⁶⁰.

Physiological signals/Biometric data. We used the E4 wristband (Empatica Inc., Milano, Italy) for physiological data acquisition. E4 is a medical-grade wearable device that offers real-time acquisition of EDA, heart rate, skin temperature, and accelerometer data from the subject’s dominant hand. EDA is measured via E4’s silver (Ag) electrode (valid range [0.01–100] μ S), while heart rate is measured via E4’s photoplethysmographic (PPG) sensor. Using Bluetooth, E4 transmits data to the subject’s smartphone. All data collected from the E4 wristband and the sampling frequencies are presented in Table 2.

The physiological signals that we measured via the E4 wristband and are provided in our dataset are as follows:

- **Heart rate:** The heart rate of a healthy individual, irrespective of gender, ranges from 60 to 100 beats per minute at rest⁶¹. However, the heart rate varies significantly with activity or emotional state⁶². A high heart rate is generally associated with stressful situations^{63,64}, but a high heart rate should not by itself be interpreted as high stress⁶⁵. Heart rate is not primary data and is generated from the BVP signal by the Empatica E4 device.
- **Skin temperature:** Skin temperature varies for various activities due to skin blood flow⁶⁶. Skin temperature ranges from 33.5 to 36.93 Celsius degrees⁶⁷. However, this can vary quite widely based on the type and length of activity and ambient temperature⁶⁸.
- **Electrodermal activity:** EDA measures the amount of sweat gland activity by calculating the skin’s electrical conductance using silver-chloride electrodes. The EDA signal is measured in units of micro-siemens (μ S). Stadler *et al.*⁶⁹, mention that EDA peaks are event-related and can be a good estimator of the body’s response to the stimulus⁷⁰. Collecting EDA signal using wristbands is not very accurate, and some artifacts can affect the data collection results. Additionally, in some subjects, the EDA signal cannot be captured, or it can be inaccurate. However, the purpose of this study is to use the data in multimodal stress detection, which remains robust in the absence of a modality or noisy data. We believe that the combination of video and physiological signals can help researchers detect stress using off-the-shelf wristbands and video, despite imperfections in any single signal stream.

Task	Action performed	Duration (min)	Expected Stress Level
T1	Reading a magazine	9	Normal
T2	Preparing a presentation	9	Stressed
T3	Delivering the presentation	5-9	Stressed
T4	Rest and recovery	9	Normal
T5	IQ test / Stroop Color-Word Test	9	Stressed
T6	Listening to calm music	9	Normal
T7	Watching amusing video(s)	9	Amused
T8	Controlled breathing exercise	9	Normal
T9	Recovery	9	Normal

Table 3. List of tasks performed by participants, total duration of the data included in the release, and expected stress level for each task.

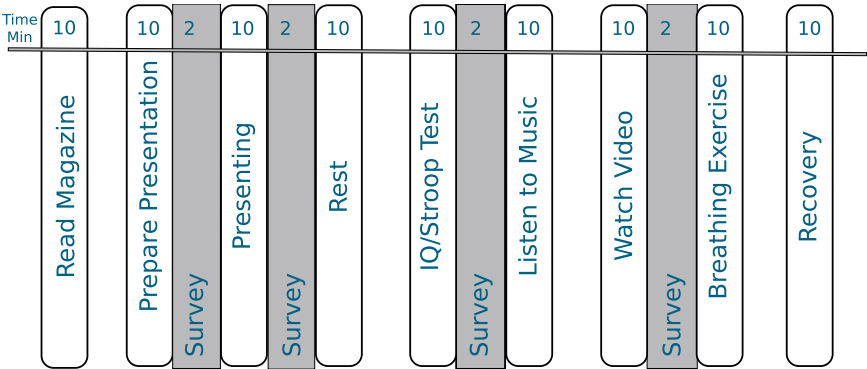


Fig. 4 Schematic overview of the EmpathicSchool data collection protocol, showing the sequence and duration of each experimental phase. We do not release the data from the gray sessions.

- **Accelerometer data:** Accelerometer sensors can be used for multiple tasks (e.g., human action recognition and step counting)⁷¹. By measuring orientation and acceleration force, the accelerometer sensor can determine the device's orientation (horizontal or vertical) and the type of movement⁷². The accelerometers vary in type (digital and analog), sensitivity, and number of axes. Our dataset provides three-axis accelerometer data that measures the orientation of the sensor in three dimensions, which enables capturing the activity more precisely.
- **Blood Volume Pulse:** Blood volume pulse is measured using a PPG that measures light absorption to find pulse volume peaks⁷³. Heart rate is calculated by analyzing the time intervals between consecutive peaks. Thus, the BVP signal provides valuable information about heart activity.
- **Inter Beat Interval:** The inter-beat interval is the time difference between two consecutive peaks in seconds, which can provide information about heart activity, such as heart rate. IBI is not primary data and is generated from the BVP signal by the Empatica E4 device.

Data collection protocol. Our stress detection study was conducted in a laboratory setting. We aimed to detect students' stress during their typical activities in a laboratory setting, while sitting behind a computer wearing a wristband. We collected data from 30 participants under different stress levels (normal and high stress). The normal stress level was assumed when the participants were at rest or performing a task that did not require significant mental effort. For the stressful tasks, we asked participants to accomplish the given assignment in a limited time⁴³. Each of these tasks was supposed to be performed for 10 minutes. However, 30 seconds of data was trimmed from the beginning and end of each task to synchronize the video and physiological signals. This gives us 9 minutes of data for each participant and each task. The actions for nine different data collection sessions (tasks) and the duration of each session are provided in Table 3. All subjects used the complete allotted time for all the tasks, except eight out of the 30 participants who completed delivering the presentation task (T3) earlier than the allotted time. Table 3 gives the expected stress level for each task.

In task T1, the students in the United States were asked to find and read an arbitrary article from Ars Technica or Wired Internet magazines, while at the Finland site, the students were asked to read an arbitrary magazine from the Internet according to their choice to reduce their stress level to achieve a baseline level of stress⁷⁴. In task T2, the participants were asked to prepare for a presentation in a limited time, followed by the presentation task T3 in the allocated time. We considered both tasks (preparing and delivering presentations) as stressful tasks. In the next task (T4), participants were asked to take a rest. In task T5, the Stroop Color-Word (Finland) or IQ test (United States) was taken by each participant to observe their stress level while taking tests in a limited time period. In task T6, we asked the participants to listen to calm music, and we considered them

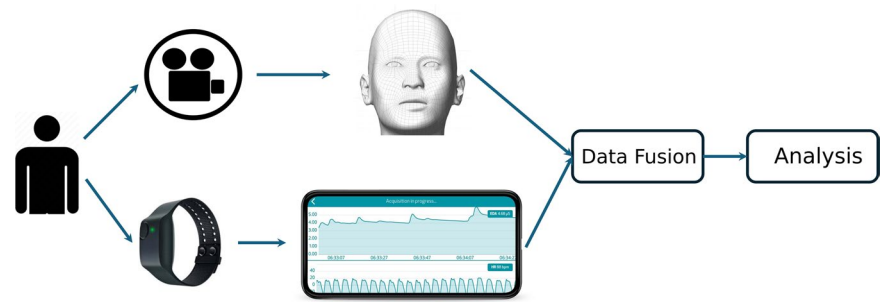


Fig. 5 Data collection apparatus.

No	Category	Question
1	Mental Demand	How mentally demanding was the task?
2	Physical Demand	How physically demanding was the task?
3	Temporal Demand	How hurried or rushed was the pace of the task?
4	Performance	How successful were you in accomplishing what you were asked to do?
5	Effort	How hard did you have to work to accomplish your level of performance?
6	Frustration	How insecure, discouraged, irritated, stressed, and annoyed were you?

Table 4. Workload questionnaire to evaluate the overall stress level of students during different tasks.

not to be stressed during this activity⁷⁵. The rest of the data collection included watching an amusing video (T7) followed by one or two different rest activities, namely controlled breathing exercises (T8) and a recovery period (T9). Only six (S1-S6) out of 30 participants were asked to have a recovery task (T9) after a controlled breathing exercise. The video and wristband data have different start and end times, so we trimmed 60 seconds of each task for data length consistency after time synchronization (30 seconds from the beginning and the end). In our study, there is a potential for the order effect. We could not mitigate it by testing different task orders, as we lacked a subject pool of sufficient size to have statistical significance. However, we sought to minimize the order effect by introducing rest periods between the tasks as illustrated in Fig. 4.

Our protocol includes three stressful tasks that induce cognitive load and social-evaluative threat, identified as two dimensions of acute stress in the stress-induction literature^{38,76}. The participants prepared for a presentation under strict time pressure (T2), a manipulation that elicits moderate cognitive load and mirrors the preparation phase of the Trier Social Stress Test (TSST)³⁸. They then delivered this talk to a camera while answering questions (T3), recreating the speech-and-arithmetic blocks of the TSST and, thereby, adding a social-evaluative component^{38,77}. Pure cognitive interference was induced in a timed Stroop/IQ task (T5), which elevates heart-rate and electrodermal activity^{41,78}. To bracket these stressors with low-arousal or down-regulation conditions, we included quiet reading (T1) as a resting baseline commonly used in stress laboratories⁷⁴, listening to calm music (T6) to induce relaxation⁷⁵, and paced breathing (T8) to recruit parasympathetic activity and vagal tone⁷⁹. Collectively, this sequence reproduces the most widely used laboratory stress paradigms, such as public speaking, time-pressured mental arithmetic, and Stroop interference, and embeds them between three validated recovery tasks, enabling inter-subject comparisons across the full arousal continuum^{38,41,80}. Moreover, these tasks were related to the students’ daily routines, which we believe to have the potential to facilitate research on the stress of students while performing daily tasks.

An Empatica E4 was worn on the wrist of the dominant arm, and a camera collected facial videos during the studies. We continuously collected the students’ physiological data and facial features during the sessions. We detected stress events while monitoring the physiological signal streams of the students. Further, We asked participants S7-S30 to validate their stress level during each stressful task by completing a self-assessment stress level check for each of the 2-minute intervals, along with a NASA questionnaire (NASA-TLX can be further used as a proxy for stress along with the stress assessment forms). The participants were also asked to complete a questionnaire if they experienced stress during the normal stress tasks. Figure 5 presents the framework for the data collection. A detailed explanation of all collected data is given in the Data records Section.

Stress detection surveys. Stress levels can be determined via subjects’ self-assessment and National Aeronautics and Space Administration Task Load Index (NASA-TLX) questionnaires^{81,82}. In this study, we provided NASA-TLX forms to participants S7-S30 and asked them to complete the questionnaires after each stressful task. Each task was divided into 5 two-minute sub-intervals, except for the first and last sub-intervals that span over 90 seconds (1.5 minutes), adding up to 9 minutes for each task. Therefore, participants (S7-S30) were asked to determine their overall stress level at each sub-interval during stressful tasks. In addition, we also asked participants (S7-S30) to rate their overall workload during different stressful tasks and provided this information and labels in the dataset. Table 4 shows the NASA questionnaires that were asked to evaluate participants’ stress levels from 0 to 20 for each sub-interval. To ensure the reliability of these scales, this study calculated Cronbach’s α ⁸³ for the questionnaire scores (0.922), which shows good internal consistency.

ID/Task	T1	T2	L2	T3	L3	T4	T5	L5	T6	T7	L7	T8	T9	Video	labels	Site
S1	✓	✓	✓	✓	X	✓	✓	X	✓	✓	X	✓	✓	X	X	Finland
S2	✓	✓	X	✓	X	✓	✓	X	✓	✓	X	✓	✓	X	X	Finland
S3	✓	✓	X	✓	X	✓	✓	X	✓	✓	X	✓	✓	X	X	Finland
S4	✓	✓	X	✓	X	✓	✓	X	✓	✓	X	✓	✓	X	X	Finland
S5	✓	✓	X	✓	X	✓	✓	X	✓	✓	X	✓	✓	X	X	Finland
S6	✓	✓	X	✓	X	✓	✓	X	✓	✓	X	✓	✓	X	X	Finland
S7	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	X	✓	✓	U.S.
S8	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	X	✓	✓	U.S.
S9	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	X	✓	✓	U.S.
S10	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	X	✓	✓	U.S.
S11	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	X	X	✓	U.S.
S12	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	X	X	✓	U.S.
S13	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	X	X	✓	U.S.
S14	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	X	X	✓	U.S.
S15	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	X	X	✓	U.S.
S16	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	X	X	✓	U.S.
S17	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	X	✓	✓	U.S.
S18	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	X	X	✓	U.S.
S19	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	X	✓	✓	U.S.
S20	✓	✓	✓	✓	✓	X	X	X	X	X	X	X	X	X	✓	U.S.
S21	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	X	✓	✓	U.S.
S22	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	X	✓	✓	U.S.
S23	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	X	✓	✓	U.S.
S24	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	X	X	✓	U.S.
S25	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	X	✓	✓	U.S.
S26	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	X	X	✓	U.S.
S27	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	X	X	✓	U.S.
S28	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	X	✓	✓	U.S.
S29	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	X	✓	✓	U.S.
S30	✓	✓	✓	✓	✓	✓	X	X	X	X	X	X	X	X	✓	U.S.

Table 5. The availability of the tasks/data for each of the participants (X mark corresponds to the case where the data is not available for the participant, ✓ shows the availability of the data).

Data records

The dataset is anonymized and available on Zenodo⁸⁴. The dataset contains 30 folders (S1 to S30), one for each participant. Each folder consists of sub-folders (T1 to T9) according to the tasks performed by participants; the participants S7 to S30 did not have the recovery task T9 in their data collection. Table 5 shows the data availability for the participants in each folder. Each folder follows these naming conventions: The first letter of the filename shows the subject ID, followed by two letters showing the task number (e.g., T1). For example, LX corresponds to the validated stress levels of the task TX.

To assess stress levels during the stressful tasks, we administered self-assessment stress level questionnaires at 2-minute intervals. However, due to concerns about survey bias arising from the predictability of the questions⁸⁵ when frequently completing both the NASA-TLX and self-assessment forms, we decided to have participants complete the NASA-TLX forms only once, at the end of the task, to measure the overall workload of the task. The labels (overall and intervals) are provided in the *stress_level.csv* files. Task T3 (delivering a presentation) lasted 5–9 minutes because some participants finished the task early. In these cases, we split the task duration into 2-minute segments for labeling, resulting in fewer labeled intervals for those participants who finished early.

Data files description. We provide the files in the task sub-folders:

- **Xception.xlsx:** The Excel file contains the facial expressions deduced at frame level for each task using MiniXception⁴⁹, trained over the FER-2013 dataset⁵⁰. The details are given in Section “Facial expression recognition”. The sampling rate is 30 frames per second.
- **HR.csv:** The file contains a single column with the average heart rate. The first row is the start time of the session in UTC time-stamp. The second row is the sampling rate expressed in Hz.
- **EDA.csv:** The file contains a single column with the average electrodermal activity. The first row is the start time of the session in UTC time-stamp. The second row is the sampling rate expressed in Hz.
- **TEMP.csv:** The file contains a single column with the average skin temperature. The first row is the start time of the session in UTC time-stamp. The second row is the sampling rate expressed in Hz.

Action Unit	AU number	Number of subjects
Lip Tightener	12	17
Lip Stretcher	11	16
Mouth Stretch	15	15
Lip Depressor	8	14
Lid Droop	25	13
Nostril Compressor	24	10
Cheek Raiser	4	9

Table 6. Distribution of Facial Action Units indicative of stress among 24 subjects with self-reported stress level labels.



Fig. 6 Analysis of subjects S1-S6 physiological signals and facial expressions for tasks T1-T9.

- **Acc.csv** The file contains three columns corresponding to the x-axis, y-axis, and z-axis accelerometer data, respectively. The first row has the start times of the session in UTC time-stamp. The second row gives the sampling rates expressed in Hz.
- **V.mp4** The video file contains raw video files of participants who consented to release their videos. This file is not available for all participants.
- **Stress_level.csv** The file holds the validated stress level of the participants extracted from their answers. Each label is for a two-minute interval (represented as 0, 1, 2, and unknown, where 0 = no-stress; 1=low-stress; 2=high-stress, unknown for unknown stress level).
- **Landmarks.csv** The file contains the normalized position of the participant's 68 facial landmarks (Dlib features described in Section "Dlib features") over time. The sampling rate of the facial landmarks is 30 frames per second, which is not provided in the landmarks csv file.
- **BVP.csv** The file contains a single column with the blood volume pulse signal. The first row is the start time of the session in UTC time-stamp. The second row is the sampling rate expressed in Hz.
- **IBI.csv** The file contains two columns; the first column shows the time-stamp, and the second column contains the corresponding inter-beat interval values.

The videos and physiological signals have exact start times and can be synced by interpolation, downsampling, or upsampling data processing methods.

Technical Validation

For the analysis presented in this paper, only subjects with complete and labeled datasets were considered. As a result, subjects S1 to S6, S20, and S30 were excluded. This exclusion was necessary to ensure the reliability and validity of the results, as missing labels or incomplete datasets could introduce bias or inaccuracies in the analysis. We used multi-class macro-averaging that calculates the metric independently for each stress level (low-, medium-, and high-stress) and then takes the average, treating all classes equally.

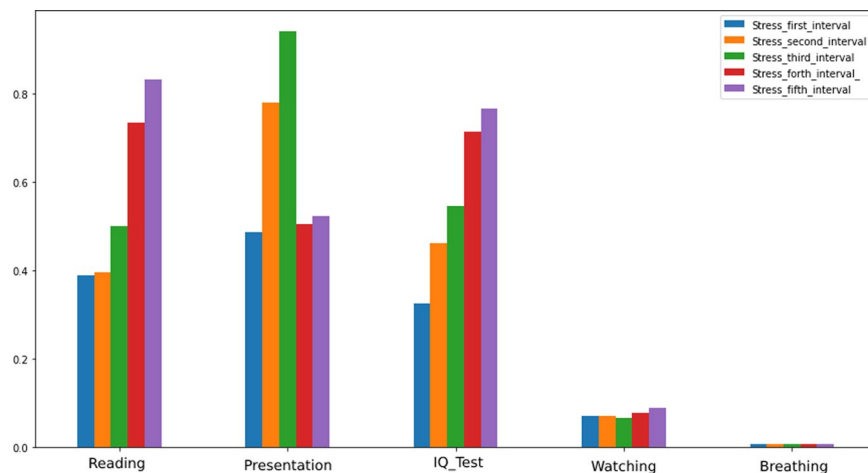


Fig. 7 Overall stress level of the students performing different stressful tasks.

Model	Features	Recall	Precision	F1_score
Random Forest				
Emotions	7	48.64	50.08	48.02
Biometric	21	65.62	64.07	61.18
Multimodal (Top 10)	10	79.72	73.46	69.32
Decision Tree				
Emotions	7	45.56	46.13	45.51
Biometric	21	58.25	58.49	56.37
Top 10	10	69.89	63.86	60.19
XGBoost				
Emotions	7	70.52	69.69	66.44
Biometric	21	79.30	74.28	70.16
Multimodal (Top 10)	10	81.64	74.82	69.28

Table 7. Performance of different models using balanced data and Leave One Subject Out.

Facial Action Units. In this paper, we investigated the correlations between stress level and frequency of the different AUs in a time window of 20 seconds among 24 subjects with self-reported stress level labels. Table 6 shows the incidence of stress manifestation through facial AUs across the subjects. We generated AUs from facial landmarks⁸⁶ and compared the scores during different stress levels. It is noteworthy that not all subjects exhibited identical AUs when exposed to stress⁸⁷. Despite this variability, we identified and reported correlations between specific AUs and stress in certain individuals, with detailed results presented in Table 6.

Machine learning and stress detection model. The EDA, HR, Temp, ACC signals, and video were collected at different sampling rates due to the variation of sensors. The frequency of the signals ranges from 1 to 30 Hz for heart rate and video data, respectively. We used the physiological signals, facial expressions, and AUs generated from videos as features for our stress detection model. We decided to use a frequency of 4 Hz after evaluating the accuracy of different models to minimize the information loss while monitoring the computational cost of the models. The code associated with the data preprocessing, upsampling, and downsampling is available on the GitHub repository⁶⁰. The machine learning classification models process the physiological signals and facial expression features. The stress detection models are trained and tested based on the participants' survey answers. The output of the stress detection models is the stress level of the participant at the mentioned time step (0: No stress, 1: medium stress, 2: high stress).

Physiological signals vs. facial expressions. We analyzed the physiological signals obtained from the Empatica E4 watch and the cla facial expression probabilities from the video data for different tasks. Since the frame rate for the video is higher than the frequency of the physiological signals, we interpolated the physiological data to make it equal in length to the length of the deduced emotions from the video frames. To smooth the emotion data (Fig. 6), we use a quadratic polynomial used in the Savitzky-Golay method⁸⁸. The smoothing filter was applied to the data at the task level by keeping the span of the smoothing filter at 30 percent. After applying the smoothing filter, we scaled the data between 0 and 1 by applying $\tilde{x}_i = \frac{x_i - x_{min}}{x_{max} - x_{min}}$, where $\tilde{x}_i$ is the scaled value. x_{min} is the minimum, and x_{max} is the maximum smoothed value at the corresponding task/session for the participant.

Figure 6 shows the happiness curves and physiological signals of subjects S1-S6 performing different tasks (T1-T9). The happiness curve is observed to fluctuate more in T5 and T7 for subjects S1-S6 as compared to other tasks. For T5, the fluctuation is attributed to the cognitive load and conflicting stimuli inherent in the task at hand, which evoke momentary frustration or challenge followed by happiness upon correct responses, resulting in emotional variability⁸⁹. The fluctuations observed during T7 likely reflect the dynamic, amusing content of the videos themselves, which may induce peaks of amusement incorporated with neutral or less engaging moments⁹⁰. For a given task, the temperature is observed to stay steady within the task; however, some differences have been observed in the temperature values between the tasks. The EDA curve is observed to stay steady during T1 and T2, while a slight variation is observed in T3. We also observed a peak heart rate value for subjects S2, S3, and S4 for T3. Although these curves illustrate how each task uniquely modulates facial expression and physiological signals, they do not establish causal relationships or explain why those modulations occur. Detailed modeling or controlled follow-up studies will be required to find the patterns of these observed fluctuations. Similar plots can be generated for comparing other emotions (Angry, Disgust, Fear, Sad, Surprise) to the physiological signals provided in the dataset.

Figure 7 shows the overall stress level of the students S7-S30 performing different tasks based on participants' feedback. For example, the overall stress level in delivering a presentation was higher than in the other tasks. The stress level of the task in the presentation had a peak exactly in the middle of the presentation, compared to reading and IQ tests, where the overall stress level of the subjects increased while reaching the end of the task. In addition, the stress level during preparation for a presentation showed similar results to those of taking a test for the participants.

Feature Extraction. A stress detection algorithm was designed to detect stress from facial expressions and physiological signals. In order to extract the features for the stress detection model, we first performed facial expression recognition in the video as described in the Section "Facial expression recognition". Consistent with prior work on wearable stress classification⁹¹⁻⁹³, we extracted features in 20-second windows with a 5-second step (75% overlap). Twenty seconds is long enough to capture several phasic EDA responses⁹⁴ and to stabilize HRV⁹⁵ statistics while remaining short enough to detect rapid stress onsets^{70,96}. In total, we used 28 features extracted from both video and physiological signals.

- **Video signal features:** The mean of the facial expressions extracted from the video was concatenated with the features extracted from biometric signals. For each of the 3 physiological signals, we calculated the mean, minimum, maximum, and standard deviation along the 20-second sliding window.
- **Physiological signals features:** In addition, we extracted the kurtosis, skewness, number of peaks, amplitude, and duration of the peak for the EDA signal. For HR, we extracted the root mean square (quadratic mean), number of peaks, amplitude, and duration of the peaks. These statistical features for physiological signals were extracted based on an earlier work by⁹⁷.
- **Labels:** The labels for each window were calculated based on the average stress values during the time window: "no stress" when $S \leq 0.65$, "medium stress" when $0.65 < S \leq 1.3$, and "high stress" when $S > 1.3$. The source code for the stress detection algorithm comprises of feature extraction, stress detection, and change-point detection, and is provided in the GitHub repository⁶⁰.

Three models (Random Forest, Decision Tree, and XGBoost) were trained from the features extracted in the second step to predict stress during the 20-second window. To evaluate the effectiveness of the model and avoid overfitting, we followed a leave-one-subject-out strategy on the test dataset, where the model is trained on data from several participants and evaluated against the remainder. This process is repeated for all the participants. This dataset is imbalanced, and stress levels are distributed as: 1- low stress: 76%, 2- medium stress: 16%, and 3- high stress: 8%. We employed the SMOTE oversampling approach to balance the data⁹⁸. The precision, recall, and F-score of these three approaches are presented in Table 7.

The emotions model used the mean of the facial expressions extracted from the video to detect stress. The biometrics model used the 21 biometric features to train the machine learning model. A large number of features may lead to an overfitted model that is not generalizable to unseen data (curse of dimensionality)⁹⁹. To reduce the number of features, we used Pearson correlation analysis¹⁰⁰ to select the top 10 features correlated with stress. The top 10 features we observed were 'lip_pucker_mean', 'lower_lip_depressor_mean', 'temp_mean', 'temp_max', 'temp_min', 'eda_mean', 'eda_max', 'eda_min', 'eda_peak_amplitude', and 'hr_min' using information gain. We trained each machine learning model with these 10 features to predict the value of stress. The code for training the machine learning models and evaluating the results is available in GitHub⁶⁰.

We evaluated the performance of Random Forest, Decision Tree, and XGBoost, using balanced data and a Leave-One-Subject-Out (LOSO) cross-validation. The models were trained using facial expression, physiological signals, and a multimodal combination of the top ten features. The Random Forest classifier, with 100 trees ($n_estimators=100$), a depth of 7 ($max_depth=7$), and a minimum sample leaf size of 5, achieved the best performance with the multimodal feature set, with a recall of 79.72%, precision of 73.46%, and an F1-score of 69.32% compared to emotions alone with a recall of 48.64% and an F1-score of 48.02%. The Decision Tree classifier, using a maximum depth of 5 and a minimum sample leaf size of 3, with the multimodal features, achieved a recall of 69.89%, precision of 63.86%, and an F1-score of 60.19%, while its performance with emotion signals alone remained limited, with a recall of 45.56% and an F1-score of 45.51%. The XGBoost model, with a maximum depth of 6 and a gamma value of 0, outperformed other models with the biometric feature set, achieving a recall of 79.30, a precision of 74.28, and an F1-score of 70.16.

Feature	Coef.	Std.Err.	Z	P> Z	[0.025	0.975]
hr_std	−0.111	0.035	−3.163	0.002	−0.180	−0.042
hr_peak_amplitude	0.038	0.017	2.299	0.021	0.006	0.071
hr_peak_duration	−0.020	0.007	−2.855	0.004	−0.034	−0.006
eda_peak_duration	0.007	0.003	2.249	0.025	0.001	0.013
inner_brow_raiser_mean	−0.058	0.018	−3.216	0.001	−0.093	−0.023
inner_brow_raiser_max	−0.046	0.017	−2.725	0.006	−0.079	−0.013
inner_brow_raiser_kurtosis	0.007	0.003	2.053	0.040	0.000	0.013
inner_brow_raiser_skewness	0.007	0.004	1.968	0.049	0.000	0.015
brow_lowerer_mean	−0.060	0.029	−2.095	0.036	−0.117	−0.004
upper_lid_raiser_min	0.076	0.038	2.006	0.045	0.002	0.151
upper_lid_raiser_max	−0.091	0.036	−2.555	0.011	−0.160	−0.021
upper_lid_raiser_std	0.051	0.022	2.272	0.023	0.007	0.095
upper_lid_raiser_skewness	0.020	0.010	2.073	0.038	0.001	0.039
cheek_raiser_feature_min	0.061	0.026	2.362	0.018	0.010	0.111
lid_tightener_mean	0.141	0.055	2.583	0.010	0.034	0.249
lid_tightener_std	−0.066	0.021	−3.058	0.002	−0.108	−0.024
nose_wrinkler_std	−0.033	0.012	−2.719	0.007	−0.057	−0.009
nose_wrinkler_kurtosis	0.008	0.004	1.982	0.047	0.000	0.017
upper_lip_raiser_min	−0.031	0.013	−2.376	0.018	−0.056	−0.005
upper_lip_raiser_kurtosis	−0.012	0.004	−3.291	0.001	−0.019	−0.005
nasolabial_furrow_deepener_mean	0.049	0.025	1.983	0.047	0.001	0.097
dimpler_std	0.029	0.014	2.040	0.041	0.001	0.058
lip_corner_depressor_mean	0.085	0.025	3.339	0.001	0.035	0.135
lip_corner_depressor_max	0.056	0.022	2.525	0.012	0.013	0.100
lip_corner_depressor_kurtosis	−0.011	0.003	−3.336	0.001	−0.018	−0.005
lower_lip_depressor_min	0.241	0.076	3.190	0.001	0.093	0.389
lower_lip_depressor_skewness	−0.007	0.003	−2.011	0.044	−0.014	−0.000
chin_raiser_max	−0.088	0.027	−3.255	0.001	−0.140	−0.035
chin_raiser_std	−0.048	0.014	−3.280	0.001	−0.076	−0.019
lip_pucker_kurtosis	0.009	0.004	2.093	0.036	0.001	0.017
lip_funneler_max	0.054	0.023	2.384	0.017	0.010	0.099
lip_funneler_skewness	0.013	0.005	2.612	0.009	0.003	0.023
jaw_drop_min	−0.066	0.030	−2.229	0.026	−0.124	−0.008
jaw_thrust_skewness	−0.040	0.017	−2.387	0.017	−0.073	−0.007
puff_std	−0.030	0.013	−2.382	0.017	−0.055	−0.005
tongue_bulge_std	−0.032	0.013	−2.548	0.011	−0.057	−0.007
nostril_dilator_std	0.027	0.012	2.199	0.028	0.003	0.052
lid_droop_mean	0.037	0.013	2.821	0.005	0.011	0.063

Table 8. Mixed linear models intercept results of different signals in different stress groups using lbfgs method.

Emotions, such as anger (angry), surprise, and sadness (sad), exhibit high correlation with stress^{56,101,102}. Several authors have previously discussed this relationship between heart rate, emotions, and stress^{63,65,103,104}. Earlier studies on stress detection evaluated the relationship between stress and skin temperature^{105,106}. The skin temperature drops during the onset of stress¹⁰⁷ and increases above the normal temperature after the onset^{107,108}. Therefore, skin temperature and stress are not directly correlated. However, a sudden decrease in skin temperature is a good indicator of early stress, but during later time periods, the skin temperature is higher. The machine learning models can use these complex relationships to identify high-stress levels, as represented in Table 7.

Linear Models with Mixed Effects. Linear Models with Mixed effects (LMMs) are an extension of linear models that incorporate fixed and random effects. These models are especially useful for hierarchical or nested data.

- **Mixed effects** in LMMs are analogous to the coefficients in standard linear regression models, representing the average effect of explanatory variables on the response variable.
- **Random effects** account for variations at different levels of the hierarchy, such as differences between subjects or groups. This dual structure allows LMMs to model the overall trend and the individual deviations from this trend.

To handle potential outliers, we applied user-level standardization. Table 8 presents the LMM results for physiological features and facial AUs when participants were treated as random effects. For example, the feature `hr_std` has a significant negative correlation ($p < 0.05$) to high stress, $b = -0.111$, $z = -3.163$, $p = 0.002$. Similarly, `inner_brow_raiser_mean` has a negative coefficient of -0.058 ($p = 0.001$), illustrating that higher inner brow-raiser activity is associated with low-stress levels. We also observe positive relationships for `lid_tightener_mean` (coefficient = 0.141 , $p = 0.010$) and `lip_corner_depressor_mean` (coefficient = 0.085 , $p = 0.001$) with higher stress levels. Multimodal features in the form of facial AUs and physiological signals together showed statistically significant correlations that are predictive for stress modeling.

For model fitting, we used the Limited-memory Broyden-Fletcher-Goldfarb-Shanno optimization algorithm, `method='lbfgs'`, which is quite efficient for large datasets with complex random effect structures. We also have robust maximum likelihood estimates. These findings point to LMM's utility for modeling with hierarchical data on stress, while accounting for the variation across both individual and group levels.

Limitations and future works

We collected data in laboratory conditions, and the following items are the limitations of our stress detection dataset:

- Not all of the dataset is covered by self-reported stress level labels. Unlabeled data do not necessarily imply a lack of stress; it just means that we did not decide to collect the labels in assumed low-stress tasks. However, we asked participants to let us know if they experienced the onset of stress in low-stress tasks. Except for one participant, none of the subjects experienced stressful moments in the low-stress tasks.
- The self-reported stress levels have a granularity of 2 minutes. Finer granularity can lead to recall bias¹⁰⁹.
- This dataset is primarily focused on labeling acute stress among participants and does not consider chronic stress as a factor in the participants. There is no sufficient research on chronic stress detection using physiological signals.
- We conducted a laboratory setting for stress detection, and the participants may experience some stress due to the new environment. However, we asked the participants to read magazines under task 1 to mitigate the newcomer's socialization through the Lens of Stress effect¹¹⁰.
- Our study primarily measures arousal rather than stress, as the tasks involved do not incorporate high stakes or a significant audience. Future research could benefit from incorporating these elements to elicit genuine responses to stress.

Code availability

We provide the code for deducing facial expressions from video data using the InceptionV3¹¹¹ model pre-trained on the FER-2013 dataset⁵⁰, and `dlib` features⁵⁵ that provided 68 facial landmarks on a facial image as discussed in Section “Dlib feature”. We also provide codes for analyzing (producing the plots/curves) physiological signals vs. facial expressions. The code is publicly available on GitHub⁶⁰ and the data is publicly available on Zenodo⁸⁴.

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Competing interests

The authors declare no competing interests.

Additional information

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